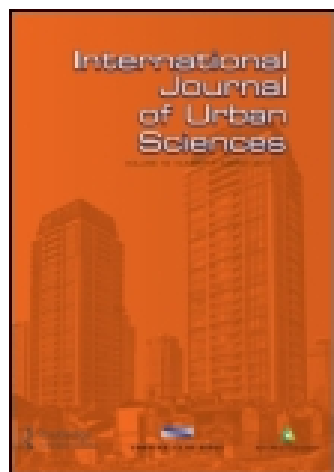


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## Delimiting boundaries of market areas of central places using the density of retail facilities in an urban space

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When Christaller's central place theory is applied to an anisotropic urban space, the hierarchical structure of the central places in hexagon shapes does not appear. Thus, we have difficulties delimiting boundaries of market areas of central places. In order to overcome this limitation, we suggest the geographic information system (GIS) method based on a hydrological modelling which uses raster density and apply the method to the retail facilities of Seoul, Korea. Kernel density function is used to measure the density of retail facilities, and search radii are set up as 500, 1000, and 5000 m by considering the size of the neighbourhood unit, the neighbourhood district, and the sub-centre region. As a result, we confirm market areas of central places with irregular shapes, which become nested in those of higher order central places. This means that central places in an urban space have the hierarchy, which is the core concept of Christaller's theory. In order to evaluate the result of delimiting market areas, we suggest the normalized internal traffic volume ratio index (NTI). The average NTIs of the market areas of central places are higher than those of administrative units, and thus, we conclude that market areas of central places are effectively delimited and the GIS method based on a hydrological modelling is useful.

**Keywords:** central place theory; boundary of market area; hydrological modelling; density of retail facilities

### 1. Introduction

In Christaller's central place theory, central places are assumed to be located at an isotropic plain having a certain population density and consumption power, and as a result, form a hierarchical structure with market areas in hexagon shapes (Christaller, 1966). For an urban space, the isotropic plain assumption cannot be adequate due to traffic convenience, regional preference, and the unevenness of urban development in addition to the geographical elements. Thus, a hierarchical structure with hexagon market areas does not appear. However, despite this limitation, the concept of hierarchical distribution of the central places is still valuable when used to explain the market structure of a large city. Thus, there is a need to develop a methodology to delimit market areas of central places in irregular shapes.

Recently, as the business geographic information system (GIS) becomes vitalized, it is becoming more general to use GIS for a firm's trade area analysis (Lee, 2006). A trade area is a geographic zone containing the people who are likely to purchase goods or services and is generally called a market area (Huff & Batsell, 1977). Thus, GIS may be used to

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analyse the market area of a central place. When examining the existing studies, there was a trend of perceiving locations based on administrative units or limiting statistical data to administrative units (Lee & Kim, 2008). In the case of Korea, Gus and Dongs were generally used as basic administrative units. However, when analysing data on the administrative units, roads that are not boundaries of the market areas may be perceived as boundaries. Overcoming this unreasonable space division needs to use point features which have the coordinates using address geocoding, and an analysis method based on their density. Based on this perception, this article aims to suggest the GIS method based on hydrological modelling using raster density and delimit boundaries of market areas of central places in Seoul, Korea. This GIS method uses the same algorithm as dividing basin boundaries using the geographical gradient in the hydrological analysis.

## 2. Literature review

### 2.1. Central place theory

The central place theory, which was created by the German geographer Walter Christaller, is simple and useful in explaining the principle of forming urban space structure (South & Boots, 1999). According to Christaller's (1966) theory, a central place is a concentrated area formed when stores providing goods and services gather in a particular area. Goods and services are classified by order; convenience goods are lower order goods and shopping goods are higher order goods.<sup>1</sup> If the orders of goods and services provided to a hinterland are high, a central place becomes a higher order place. Higher order central places nest lower order and thus a hierarchical structure appears. A central place exerts influence in a hexagon shape called a market area, and differs by size, which depends on the order of the central place.

Applying Christaller's theory in an urban space was first attempted by Carol (1960), who studied the city of Zürich, and subsequently, various researches have been conducted on many cities around the world. The previous studies (Clark & Rushton, 1970; Davies, Holz, & Robertus, 1999; Lee, 1992) generally show that it is difficult for market areas in hexagon shapes to be formed in an urban space.

The hierarchical structure of central places in Christaller's theory is based on overly fixed assumptions and thus shows an unrealistic symmetric structure (Berry, Parr, Epstein, Ghosh, & Smith, 1988). Lösch alleged that there was a possibility that all of the lower order central places might not be nested inside the scope of influence of the higher order central places (Lösch, 1954; Sung, 2004). In terms of discordance between central place theory and reality, a considerable amount of research modifying Christaller's theory has appeared in the geographical literature (Berry et al., 1988; South & Boots, 1999). One example is the research that analyses the effect of a non-uniform population density or demand surface (Beavon, 1974; Gusein-Zade, 1993; Rushton, 1971). A second example is the literature that modifies the assumptions relating to spatial behaviour by incorporating multi-purpose shopping trips (Eaton & Lipsey, 1982; Keane 1989; Mulligan, 1984).

Most of previous studies have focused on the location of central places and a hierarchical structure; however, boundaries of market areas have seldom been considered to be important. Some studies delineate market area of central places but they use administrative units or official statistical units (Lee, 1992). In this case, boundaries of the market areas follow administrative or statistical units, and thus they cannot be detailed and adequate. Overcoming this limitation needs the GIS method based on point features that have the coordinates using address geocoding (Lee, 2006).

## 2.2. Trade area analysis using GIS

Apart from Christaller's theory, which was developed in Germany, trade area researches have been conducted mainly in the USA, and various models were developed. Reilly's (1931) model, Converse's (1949) break-point model, and Huff's (1963) model are representative models, which are based on the gravity model. Delimiting boundaries of trade areas is an important issue and a considerable amount of research has appeared (Bucklin, 1971; Current & Storbeck, 1994; Huff & Batsell, 1977; Huff & Rust, 1984; Peterson, 1974). However, it has only been limitedly used because plenty of manpower and time were required to actually apply these models to the field. Such limitations in use have been improved along with using GIS, especially the business GIS. The main data of a legacy GIS are mostly digital maps. However, the digital maps in the business GIS are used for the purpose of providing information on nearby situations to users simply as a background, and the data under actual interest are dynamic data, such as geo-demographics or customer data. Such dynamic data are mostly used as point features which have the coordinates using address geocoding (Lee, 2006).<sup>2</sup>

The method of finding locations where economic activities are concentrated is conducted mostly by using statistical data when finding trade areas. To accomplish this, various forms of statistical data that basically use administrative units are used. In addition, a more detailed analysis is available because the statistical information by Statistics Korea is distributed by the census tracts. However, there is a problem that the statistics may be distorted as the size and shape of census tracts are uneven.<sup>3</sup> Thus, the method of analysing trade areas based on a raster density is often used these days. Conducting the method requires raw data on economic activities, such as the location of workplaces, business types, location of the customers, and the size of economic activities, and therefore such method has been mostly used in an individual firm's trade area analysis or used on an experimental level.

There are largely two methods of delimiting boundaries of trade areas: the method of using customer information and the method of not using customer information. The method of using customer information uses the customer spotting technique that mostly divides trade areas using the actual location of the customers. However, this method incurs the burden of having to investigate customer information first, and trade areas may be overlapped or areas that do not belong to any trade areas may appear. As the method of not using customer information, the buffer analysis using a certain distance or the range analysis using a certain travel time through road networks has been used when determining the trade area (Lee, Hong, & Park, 2010; Shin, Jung, & Lee, 2002). However, overlapped trade areas and omitted areas still appear. As an alternative method, the Thiessen polygon is widely used these days<sup>4</sup> but has a limitation not to adequately reflect actual space phenomenon because the Thiessen polygon is based on the evenness of space. Some previous researches used the GIS method based on a raster density analysis (Lee & Kim, 2008; Shin, Jung, & Lee, 2002). They analysed effectively individual trade areas; however, they did not consider influences of individual trade areas on the entire market.

## 3. Research method

### 3.1. Hydrological analysis: delineating the catchment boundaries

In geography, catchment is the process of collecting water, in particular the process of water flowing from the ground and collecting in a river or a lake. The delineation of catchment boundaries is necessary when modelling a hydrological problem. Drainage boundaries are used in water budgeting, water quality evaluation, as well as many other engineering

applications. Therefore, accurate delineation of drainage boundaries is very important in modelling process.

Manual delineation of boundaries from topographic maps can be done with contour lines. But the manual operation is a difficult process, and slight differences between operators are unavoidable. On the contrary, catchments delimited from digital elevation model (DEM) will be consistent, regardless of the user. Only the accuracy and resolution of DEM will affect the accuracy of catchment boundaries.

DEM is a digital record set of terrain elevations for ground locations of grids, which are divided pieces of ground with cells of equal size. Elevation information is derived from standard topographic maps or other surveying methods. DEM data are easily processed with the GIS software with raster functionalities, such as ArcGIS with Spatial Analyst extension. The general procedures of delineating the catchment area are composed of the following.

#### 3.1.1. *Flow direction*

Flow direction of every grid cell is calculated from the DEM using the gradient vector of the grid cell. In DEM, the elevation variation has resulted in slope and aspect, components of the gradient vector.

#### 3.1.2. *Flow accumulation*

The flow direction data have resulted in a flow accumulation within the catchment. If it rains, raindrops will be accumulated according to the flow direction of every grid cell. Accumulated grid values are the number of cells that drain to an individual cell.

#### 3.1.3. *Stream definition and stream segmentation*

Using a flow accumulation grid, streams are defined by a threshold flow accumulation value. The cells with the value greater than threshold are coded as streams. Then the stream network is divided into individual stream segments.

#### 3.1.4. *Catchment delineation*

A catchment is the land area that drains to a given stream. Any raindrops falling on a catchment has a unique path to a stream. So catchment boundaries can be delineated from the segmented stream following the opposite of flow direction.

### 3.2. *Analogy*

In hydrological modelling, catchments are divided by using the principle that raindrops flow from a higher spot to a lower spot following the flow direction. This is similar to the traffic concentration to a central place with high density from nearby areas. Actually the term of catchment area is used in the service industry as well as in hydrological modelling. The catchment area of a school, hospital, or other service area means the area that it serves. In hydrologic analysis, gradient of elevation differences becomes flow direction, and in market area analysis, gradient of business density differences becomes traffic direction. Therefore, in this study we have proposed the analogous method to the hydrological modelling for delineating market boundaries (Figure 1).

As shown in Figure 1, applying the analogous method to delineate the market area, density analysis of the business data is necessary. Locations of the business facilities can be

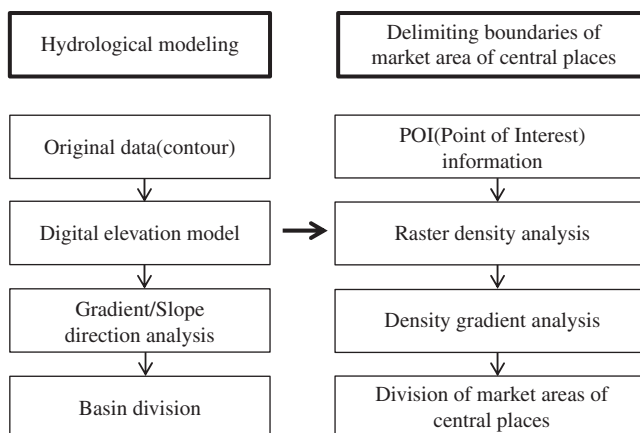


Figure 1. Analogy of the hydrological modelling for delimiting market area boundaries.

compiled with point of interest of the navigation database or the yellow pages of the phone book. Business density will be calculated using the point features of the business facilities in GIS. In the point density analysis as such, the kernel density is known to be adequate (Lee & Kim, 2008).

The kernel density is calculated by the number of point features included within a search radius from every raster grid cell. Here, points nearby the cell are given a high weight value and points that are far from the cell are given a low weight value. Kernel density estimation has an advantage because only search radius and weighting function will affect the result. Also, a large search radius helps in determining the global trend, and the details can be grasped using a short search radius. Inverse quadratic function is used as weighting function, in general (Borruso, 2003; Thurstain-Goodwin & Unwin, 2000).

Meanwhile, the gradient in hydrology is defined from a high position to a low position, but the gradient in market area delineation should be defined from a low density to a high density. So, subtracting the calculated density from the maximum value among the calculated density values, the area with the high density becomes the area with the low elevation, and the area with the low density becomes the area with the high elevation, thus, a retail business density DEM may be created for market area division (Figure 2). The location with the local maximum density in the retail business density DEM becomes the central place. A gradient where a density decreases appears as getting farther away from the central place. By using this gradient, it is possible to extract a market area by applying the same algorithm as dividing the basin boundary in the hydrological analysis. Once the density DEM for the subject area is created, the basin may be divided using the hydrological analysis function of the GIS software, and the divided basins each correspond to a market area existing in the subject area. Moreover, when delimiting market area boundaries, more and more market areas with high centrality will appear by expanding the search radius necessary in the density analysis to a wider scope, which assist a researcher in finding the hierarchy between market areas.

The divided market area can be classified into an area formed where a large number of retail facilities are densely located and an area forming an independent market area with a small number, and the former can be further classified into market areas connected with nearby areas and market areas where a narrow space shows high density by being isolated from nearby areas. Thus, it is not possible to find the hierarchy of market areas only with

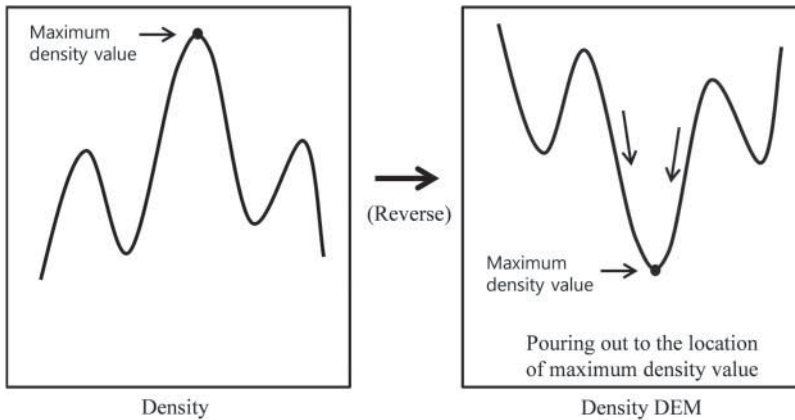


Figure 2. The concept of density DEM.

the high–low density, but the hierarchy of market areas can be found by understanding the scale of market areas by figuring out the size of the divided market area and the number of retail facilities located within the area.

## 4. Results of analysis

### 4.1. Data

The spatial scope of this research is Seoul, Korea. To analyse boundaries and hierarchy of central places, retail business found under the business categories from February 2007 phone book was used. The data used in this research are the point feature data containing the coordinates and business type of each retail facility (Figure 3). Geocoding is performed using the address from the information included in the phone book. Business types of phone book data are composed of 14 categories and 70 sub-categories. In this study, 10 sub-categories, the subjects of daily consumption, are selected for analysing kernel density. The selected categories are seven sub-categories in retail category, business-related service, daily life service, and food service (Table 1). Subsequently, 166,493 phone numbers are used, whereas 642,636 data for Seoul.

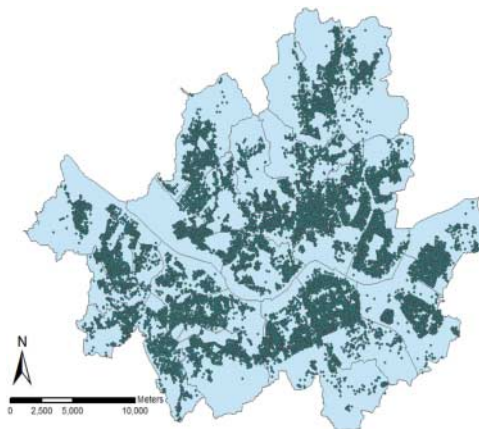


Figure 3. Retail facilities in Seoul.

Table 1. Description of selected data in a phone book.

| Category                         | Sub-category                                                 | Count   | Selected |
|----------------------------------|--------------------------------------------------------------|---------|----------|
| Natural resources                | Agriculture and livestock and four kinds of other            | 0       | 0        |
| End goods manufacturing          | Food manufacturing and eight kinds of other                  | 26,505  | 0        |
| Intermediate goods manufacturing | Textile manufacturing and seven kinds of other               | 15,573  | 0        |
| Machinery                        | Machine and equipment manufacturing and seven kinds of other | 32,163  | 0        |
| Power, gas, and water            | Power and gas and one kind of other                          | 2364    | 0        |
| Construction                     | General construction and one kind of other                   | 32,415  | 0        |
| Wholesale trade                  | Natural resources wholesale and four kinds of other          | 79,387  | 0        |
| Retail business                  | General retail                                               | 6778    | 6778     |
|                                  | Food and cigarette retail                                    | 7354    | 7354     |
|                                  | Apparel and accessory retail                                 | 10,309  | 10,309   |
|                                  | Dwelling goods retail                                        | 14,904  | 14,904   |
|                                  | Cultural goods retail                                        | 18,380  | 18,380   |
|                                  | Automobile-related retail                                    | 11,031  | 11,031   |
|                                  | Other retail                                                 | 11,634  | 11,634   |
| Transportation                   | Land transportation and two kinds of other                   | 31,030  | 0        |
| Information service              | Information/communication and four kinds of other            | 70,916  | 0        |
| Financial business               | Banking and two kinds of other                               | 39,002  | 0        |
| Real estate                      | Real estate and one kind of other                            | 33,603  | 0        |
| Service                          | Business-related service                                     | 43,291  | 43,291   |
|                                  | Daily life service                                           | 8383    | 8383     |
|                                  | Food service                                                 | 34,429  | 34,429   |
|                                  | Medical service                                              | 17,380  | 0        |
|                                  | Education and cultural service                               | 37,137  | 0        |
|                                  | Travel and lodging service                                   | 11,816  | 0        |
|                                  | Sports and entertainment service                             | 5212    | 0        |
|                                  | Social service                                               | 4187    | 0        |
| Public and NGO                   | Public administration and two kinds of other                 | 25,589  | 0        |
| Unclassified                     |                                                              | 11,864  | 0        |
| Sum                              |                                                              | 642,636 | 166,493  |

#### 4.2. Division of market areas

The kernel density function included in the Spatial Analyst Tools of ArcGIS 10 was used to calculate the density of retail business as the first step of the analysis. The pixel size of the kernel density function was set at 100 m, and the difference occurring to the results regarding market area division depending on the hierarchy of market area was examined by changing the search radius to 500, 1000, and 5000 m. The 500 m search radius is determined by considering the radius of Perry's neighbourhood unit, which is generally perceived as a basic life zone, and also the radius of transit-oriented development. The 1000 m search radius is determined by considering a distance between two subway stations, and a neighbourhood district, which means walking zone. The 5000 m search radius was suggested by considering the size of a sub-centre's hinterland and a large life zone. Seoul is divided as five large

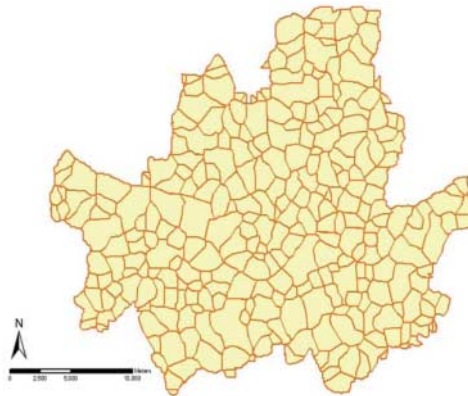


Figure 4. Boundaries of market areas by the 500 m search radius.



Figure 5. Boundaries of market areas by the 1000 m search radius.

life zones in 2020 comprehensive plan: downtown zone, northeast zone, northwest zone, southeast zone, and southwest zone. The 5000 m search radius is determined by considering the radii of large life zones measured from 4.2 to 7.4 km.

Next, the density DEM was created and the hydrological analysis was implemented by using the ArcHydro Tool. The flow direction and the flow accumulation were sought from the density DEM and the entire Seoul area was divided into independent market areas by dividing the areas into basins.

As a result of the analysis, Figures 4–6 show the boundary of the market areas when expanding the search radius to 500, 1000, and 5000 m. By using the 500 m search radius as the neighbourhood unit size, the market area was divided into 296 areas as in Figure 4. After using the 1000 m search radius as the neighbourhood district size, the market area was divided into 103 areas as in Figure 5, and lastly, after using the 5000 m search radius as the sub-centre region size, the market area was divided into 4 areas as shown in Figure 6. When changing the search radii and overlapping market areas, the lower order market areas appeared to be nested in a higher order market area as in Figure 7. This shows the hierarchy of central places, and explains how the market area of higher order central place is divided and includes the market areas of lower order central places. Table 2 shows the result of market area analysis using 500, 1000, and 5000 m. The number of market areas decreased

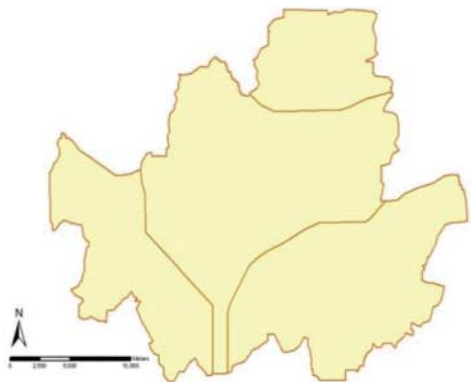


Figure 6. Boundaries of market areas by the 5000 m search radius.



Figure 7. Overlapping boundaries of market areas.

Table 2. The analysis result of the market areas of the central places.

|                                            | Neighbourhood unit | Neighbourhood district | Sub-centre region |
|--------------------------------------------|--------------------|------------------------|-------------------|
| Search radius                              | 500 m              | 1000 m                 | 5000 m            |
| Number of market areas                     | 296                | 103                    | 4                 |
| Range of market areas (km <sup>2</sup> )   | 0.18–8.55          | 0.24–17.42             | 70.70–237.76      |
| Average of market areas (km <sup>2</sup> ) | 1.91               | 6.59                   | 148.20            |

and the size of market area increased as the search radius expands. In other words, the market area divided by the 500 m search radius appeared close to the neighbourhood unit, thus, marking the boundary of lower order central place market areas, and the size increased when getting closer to the higher order central place market area. The market area divided in this way corresponds to the minimum level of which a hierarchical central place can economically survive, in other words, to the spatial scope of having the demand level, the number of customers, or the number of competitors that allow maintaining the function of a central place.

#### 4.3. Adequacy evaluation for market area division

Until now, there were no evaluation methods for assessing the appropriateness of the delineated market area, as well as the delineating method of exclusive and exhaustive market areas. In this research, we have proposed the verification method for assessing the appropriateness of market area delineating method.

Delineated market area assumes that the traffic originated from the low density areas to high density central area caused by the differences in density. In other words, traffic for shopping is generated from surrounding areas with low retail density to central place. Accordingly, it is expected that the internal shopping traffic within the market area is greater than the external shopping traffic between the market areas. Under this assumption, we will assess the appropriateness of the delineated market area, comparing internal traffic volume with external traffic volume.

The basic traffic volume data provided by the Metropolitan Transportation Authority were used for the analysis. These data provide only origin-destination (OD) information in administrative units; thus, it has limitations in providing an accurate evaluation. In spite of the limitations, these data are used for the assessment because they provide the most detailed OD information by the purpose of trip. The purposes of trips are categorized in eight ways: home-based work trip, home-based school trip, home-based private educational trip, home-based shopping trip, home-based other trip, non-home-based business trip, non-home-based shopping trip, and non-home other trip. In this research, only two types of shopping trips, home-based shopping trip and non-home-based shopping trip, are used for the analysis.

The OD information constructed per administrative unit was analysed by overlapping with the divided market area in order to construct OD data in the market area unit. The market area polygon divides the administrative unit, and the divided OD of the administrative unit was combined into the OD data for the market area. The entire traffic volume of the market area is the sum of all the shopping trips from and to the market area. The internal traffic volume within the market area means the shopping trip had the same starting and ending point. Then, the ratio of internal traffic volume to entire traffic volume was calculated. However, in this case, because the ratio of short-distance traffic volume is relatively higher than long-distance traffic volume, the ratio of internal traffic volume to entire traffic volume increased as the area size of the subject became larger, thus, the ratio of internal traffic volume naturally increased as well when largely dividing the market area size. In order to reduce the effect by area size, this research has calculated the normalized index on the ratio of internal traffic volume of each market area by using

$$NTI_i = \frac{RITV_i / A_i}{RITV_{Seoul} / A_{Seoul}}. \quad (1)$$

Here,  $NTI_i$  is the normalized internal traffic volume ratio index of market area  $i$ ,  $RITV_i$  is the ratio of internal traffic volume to entire traffic volume of market area  $i$ ,  $A_i$  is the size of market area  $i$ ,  $RITV_{Seoul}$  is the ratio of internal traffic volume to the entire traffic volume of Seoul, and  $A_{Seoul}$  is the size of Seoul.

Table 3 shows the result of calculating the normalized internal traffic volume ratio indexes (NTIs), and the average of administrative units, Gus appeared to be 1.04, while the average of administrative units, Dongs appeared to be 1.61. The NTI average of Gus, which has a larger area size, appeared to be close to 1, which is a similar level as the entire value of Seoul. On the other hand, the average NTIs of the 1000 m search radius and 500 m search radius market areas appeared, respectively, to be 2.23 and 2.17, a high value regardless of area

Table 3. NTIs of internal traffic volume.

|                                 | Administrative<br>unit (Gus) | Administrative<br>unit (Dongs) | Market area<br>(S.R. = 1000 m) | Market area<br>(S.R. = 500 m) |
|---------------------------------|------------------------------|--------------------------------|--------------------------------|-------------------------------|
| Average area (km <sup>2</sup> ) | 24.28                        | 1.43                           | 6.59                           | 1.91                          |
| Average of NTIs                 | 1.04                         | 1.61                           | 2.23                           | 2.17                          |
| Standard deviation              | 0.26                         | 0.93                           | 1.26                           | 1.52                          |
| 1 or more (a)                   | 13                           | 313                            | 69                             | 162                           |
| Under 1 (b)                     | 12                           | 111                            | 16                             | 48                            |
| Max                             | 1.54                         | 6.86                           | 7.06                           | 8.79                          |
| Min                             | 0.59                         | 0.02                           | 0.08                           | 0.09                          |
| a/b                             | 1.08                         | 2.82                           | 4.31                           | 3.38                          |

Notes: S.R. refers to the search radius. S.R. = 1000 m and S.R. = 500 m correspond to those of Table 2.

size. This means that this research accurately delineates the market areas where the internal traffic ratio is high. This can also be shown from the fact that ratios of the number of market areas with 1 or more in NTI compared to market areas with less than 1 appear to be high in the 1000 m search radius and the 500 m search radius market areas, and maximum values of the NTIs appear to be higher than those of the administrative units, Gus and Dongs.

## 5. Conclusions

The hierarchical structure of central places with the market areas in hexagon shapes alleged by Christaller's central place theory does not appear in an anisotropic urban space, therefore there is difficulty in delimiting boundaries of the market areas. In order to overcome this limitation, this research suggests the GIS method based on a hydrological modelling using raster density and applies the method to the retail facilities of Seoul, Korea.

Measuring density is obtained by kernel density function, and search radii are set up as 500, 1000, and 5000 m by considering the size of the neighbourhood unit, the neighbourhood district, and the sub-centre region. As a result, market areas in irregular shapes are delimited and when expanding the search radius to 500, 1000, and 5000 m, the size of market areas of central places gradually increases and the market areas of lower order central places appear to be nested in those of higher order. This can be considered as the actual appearance of the hierarchy, the core concept of the central place theory, in an urban area. To evaluate the adequacy of market area division, the NTI based on OD trips is suggested. The averages of NTIs of market areas are higher than those of the administrative units, Gus and Dongs. This result means that the market area division is more effective than the market division in administrative units by existing researches.

In terms of methodology, the GIS method based on a hydrological modelling has never been used for market area analysis and thus may contribute to the development of market area studies. However, accurate boundaries of market areas could not be found in this research because we only conducted a density analysis using the location of retail facilities based on a phone book without considering size. This requires more detailed research in the future.

## Notes

1. A convenience good is an inexpensive product that is easily purchased. Cigarettes, candies, and soft drinks are examples. A shopping good is a high-end product occasionally bought by consumers that are generally compared for their quality, cost, and features.

2. The point feature is a geographical element having the location of an XY coordinate, which is used when designating locations on a map by using the addresses of buildings or businesses occupying an actual space (Lee, 2006).
3. The Modifiable Area Unit Problem (MAUP) is a problem known to frequently occur when analysing a geographical phenomenon. The MAUP indicates problems that may occur because the regional statistics change depending on the shape and size of a sector when aggregating statistics on space (Kim, 2011).
4. Thiessen polygon is also called the Voronoi polygon and is a polygon formed from the seed points. The boundary of the Thiessen polygon is set up so that the distance to the seed point on the polygon is closer than any other seed points on nearby polygons.

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